



GOSFORD HIGH SCHOOL

2013

HSC ASSESSMENT TASK 3

EXTENSION 1 MATHEMATICS

General Instructions:

Attempt all Questions

- Working time: 60 minutes
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question in Section 2 should be Started on a separate page.
- All necessary working should be shown in every question.

Total marks: - 40

MULTIPLE CHOICE	/ 4
INVERSE FUNCTIONS	/ 24
GENERAL SOLUTIONS	/ 3
FURTHER RATES OF CHANGE	/ 9
TOTAL	/ 40



Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct
↓

Start
here

→ 1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

MULTIPLE CHOICE

(Answer on the sheet provided)

1) The exact value of $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ is

A) $-\frac{\pi}{6}$

B) $\frac{5\pi}{6}$

C) $-\frac{\pi}{3}$

D) $\frac{2\pi}{3}$

2) $f(x) = 3x + 2$ for $x \geq -2$

Which of the following represents $f^{-1}(x)$

A) $f^{-1}(x) = \frac{x-2}{3}$ for $x \geq -2$

B) $f^{-1}(x) = \frac{2-x}{3}$ for $x \geq -2$

C) $f^{-1}(x) = \frac{x-2}{3}$ for $x \geq -4$

D) $f^{-1}(x) = \frac{x-2}{3}$ for $x \leq -4$

3) If $\tan \theta = \sqrt{3}$ then

A) $\theta = 2n\pi \pm \frac{\pi}{3}$

B) $\theta = 2n\pi + \frac{\pi}{3}$

C) $\theta = n\pi \pm \frac{\pi}{3}$

D) $\theta = n\pi + \frac{\pi}{3}$

- 4) A square is expanding so that its area is increasing at a constant rate of $12\text{cm}^2/\text{sec}$. The rate of increase of the side of the square when the area is 36cm^2 is:

- A) $\frac{1}{6}\text{cm}/\text{sec}$ B) $\frac{1}{3}\text{cm}/\text{sec}$
- C) $2\text{cm}/\text{sec}$ D) $1\text{cm}/\text{sec}$
-

Question 5. (Start a new page)

- a) Draw a neat sketch of $y = 3\sin^{-1} 2x$ showing clearly the domain and range. 2

b) Differentiate

- i) $\sin^{-1} 2x$ 1
- ii) $\tan^{-1}(x+2)$ 1
- iii) $x^2 \sin^{-1} x$ 2
- iv) $x \cos^{-1} x - \sqrt{1-x^2}$ 2

c) Find the indefinite integrals of the following.

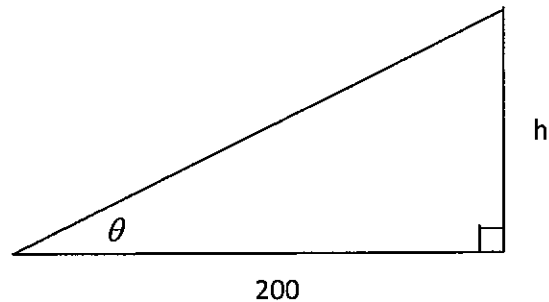
- i) $\frac{1}{\sqrt{4-x^2}}$ 1
- ii) $\frac{1}{9+x^2}$ 1
- iii) $\frac{x+1}{x^2+4}$ 2

Question 6 (Start a new page)

a) Find all solutions to $\sqrt{3} \cos x + \sin x = 1$

3

b)

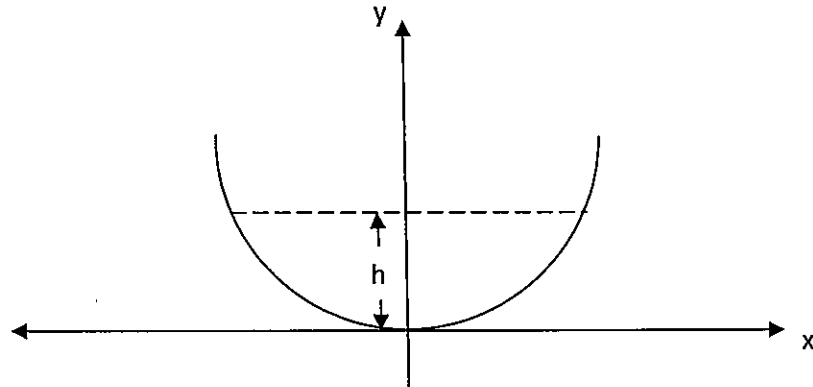


At an air show, a Harrier Jump Jet leaves the ground 200 *metres* from an observer and rises vertically at the rate of 25 *m/sec*. At what rate is the observer's angle of elevation of the aircraft changing when the jet is 500 *metres* above the ground?

3

QUESTION 6 CONTINUED OVER PAGE

c)



The diagram shows the lower half of the circle whose equation is $x^2 + y^2 - 20y = 0$. A hemispherical bowl is obtained by rotating the semi-circle about the y-axis.

- i) Show that when the depth of water in the bowl is h cm., the volume of water, V , in the bowl is given by:

$$V = \frac{\pi h^2}{3}(30 - h) \text{ cm}^3 \quad 2$$

- ii) Show that when the depth of water in the bowl is h cm., the surface area of water, S , exposed to the open air is given by:

$$S = \pi h(20 - h) \text{ cm}^2 \quad 2$$

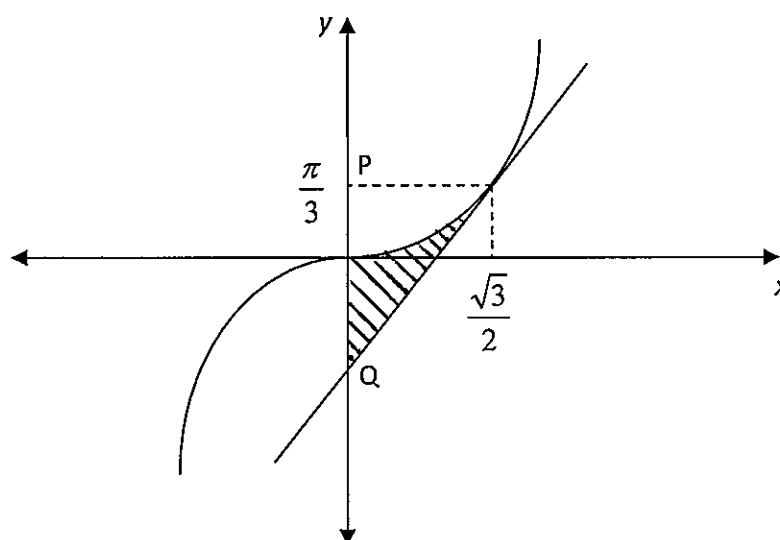
- iii) If water is poured into the bowl at the rate of $50 \text{ cm}^3 / \text{sec}$ determine, at the instance when the depth of water is 5 cm the rate of increase of the open surface of water in cm^2 / sec . 2

Question 7 (Start a new page)

a) Use the substitution $u = x^2$ to evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx$

3

b)



The diagram shows the region bounded by $y = \sin^{-1} x$, the 'y' axis and the tangent to the curve at the point $(\frac{\sqrt{3}}{2}, \frac{\pi}{3})$.

Show that the volume of the solid formed when the region is rotated about the 'y' axis is $\frac{\pi}{24}(9\sqrt{3} - 4\pi)$ cubic units.

4

c) A function is defined by $g(x) = \frac{x+2}{x+1}$, for $x < -1$

i) Find $g^{-1}(x)$ and state its domain and range.

3

ii) Find any point(s) of intersection for the curves

$y = g(x)$ and $y = g^{-1}(x)$

2

END OF THE EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

EXTENSION 1 ASSESSMENT TASK SOLUTIONS (TASK 3 2013)

MULTIPLE CHOICE

$$\begin{aligned} 1) \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= \pi - \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\ &= \pi - \frac{\pi}{6} \\ &= \frac{5\pi}{6} \quad (B) \end{aligned}$$

$$\begin{aligned} 2) f(x) &: 3x+2 \quad D: x \geq -2 \\ &\quad \therefore R: y \geq -4 \end{aligned}$$

$$\begin{aligned} f^{-1}(x): \quad x &= 3y+2 \\ y &= \frac{x-2}{3} \\ D: x &\geq -4 \end{aligned}$$

(C)

$$3) \quad (D)$$

$$4) \quad \frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$12 = 2x \cdot \frac{dx}{dt}$$

$$\frac{dx}{dt} = \frac{12}{2x}$$

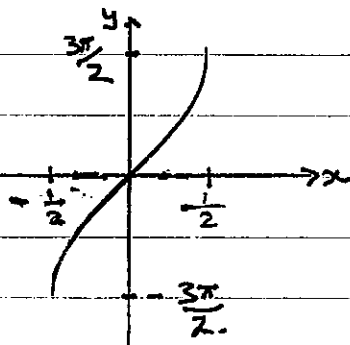
$$(Area = 36 \text{ cm}^2 \rightarrow x=6)$$

$$\frac{dx}{dt} = \frac{12}{12}$$

$$= 1 \quad (D)$$

Question 5

$$a) y = 3\sin^{-1} 2x$$



$$b) i) \frac{d}{dx}(\sin^{-1} 2x) = \frac{2}{\sqrt{1-4x^2}}$$

$$ii) \frac{d}{dx}(\tan^{-1}(x+2)) = \frac{1}{1+(x+2)^2}$$

$$\begin{aligned} iii) \frac{d}{dx}(x^2 \sin^{-1} x) \\ = \frac{x^2}{\sqrt{1-x^2}} + 2x \sin^{-1} x. \end{aligned}$$

$$\begin{aligned} iv) \frac{d}{dx}(x \cos^{-1} x - \sqrt{1-x^2}) \\ = \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x - \frac{1}{2} \left(\frac{-1}{\sqrt{1-x^2}} \right) x - 2x \\ = \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x + \frac{x}{\sqrt{1-x^2}} \\ = \cos^{-1} x. \end{aligned}$$

$$c) i) \int \frac{dx}{\sqrt{4-x^2}} = \sin^{-1}\left(\frac{x}{2}\right) + C$$

$$ii) \int \frac{dx}{9+x^2} = \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

$$\begin{aligned} iii) \int \frac{x+1}{x^2+4} dx &= \int \frac{x}{x^2+4} + \frac{1}{x^2+4} dx \\ &= \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C \end{aligned}$$

Question 6.

a) $\sqrt{3} \cos x + \sin x = 1$

Let $\sqrt{3} \cos x + \sin x = R \cos(x - \alpha)$

$\therefore \sqrt{3} \cos x + \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha$

equating coefficients

$R \cos \alpha = \sqrt{3} \quad \dots (1)$

$R \sin \alpha = 1 \quad \dots (2)$

(2) $\div$ (1) $\tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = \frac{\pi}{6}$

(1)² + (2)² $R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4$

$R^2 = 4$

$R = 2$

$\therefore \sqrt{3} \cos x + \sin x = 2 \cos(x - \frac{\pi}{6})$

$\therefore \sqrt{3} \cos x + \sin x = 1$

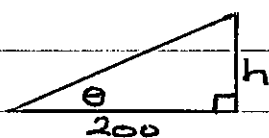
$\Rightarrow 2 \cos(x - \frac{\pi}{6}) = 1$

$\cos(x - \frac{\pi}{6}) = \frac{1}{2}$

$x - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$

$x = 2n\pi + \frac{\pi}{2} \quad \text{OR}$

$2n\pi - \frac{\pi}{6}$



$\tan \theta = \frac{h}{200}$

$\theta = \tan^{-1} \left(\frac{h}{200} \right)$

$\frac{d\theta}{dh} = \frac{\frac{1}{200}}{1 + \left(\frac{h}{200} \right)^2}$

$= \frac{200}{40000 + h^2}$

Now $\frac{d\theta}{dt} = \frac{d\theta}{dh} \times \frac{dh}{dt}$

$= \frac{200}{40000 + h^2} \times 25$

When $h = 500$

$\frac{d\theta}{dt} = \frac{200}{40000 + 250000} \times 25$

$= \frac{1}{58} \text{ } ^\circ / \text{Sec.}$

ALTERNATE METHOD

$\tan \theta = \frac{h}{200}$

$h = 200 \tan \theta$

$\frac{dh}{d\theta} = 200 \sec^2 \theta$

$\frac{dh}{dt} = \frac{dh}{d\theta} \times \frac{d\theta}{dt}$

$25 = 200 \sec^2 \theta \frac{d\theta}{dt}$

$\therefore \frac{d\theta}{dt} = \frac{25}{200 \sec^2 \theta}$

$= \frac{1}{8(1 + \tan^2 \theta)}$

$= \frac{1}{8 \left(1 + \frac{h^2}{200^2} \right)}$

$= \frac{5000}{29000}$

$= \frac{1}{58} \text{ } ^\circ / \text{Sec.}$

$$c) i) \quad x^2 + y^2 - 20y = 0$$

$$x^2 = 20y - y^2$$

$$V = \pi \int_0^h x^2 dy$$

$$= \pi \int_0^h (20y - y^2) dy$$

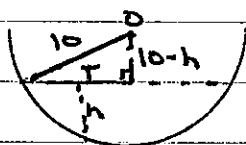
$$= \pi \left[10y^2 - \frac{y^3}{3} \right]_0^h$$

$$= \pi \left[(10h^2 - \frac{h^3}{3}) - 0 \right]$$

$$= \pi \left[\frac{30h^2 - h^3}{3} \right]$$

$$= \frac{\pi h^2}{3} (30 - h) \text{ cm}^3$$

ii)



$$\begin{cases} x^2 + y^2 - 20y + 100 = 100 \\ x^2 + (y-10)^2 = 100 \\ \therefore \text{radius} = 10 \end{cases}$$

$$r^2 = 10^2 - (10-h)^2$$

$$= 100 - 100 + 20h - h^2$$

$$= 20h - h^2$$

$$A = \pi r^2$$

$$= \pi (20h - h^2)$$

$$ii) \quad \frac{dv}{dt} = \frac{dv}{dh} \times \frac{dh}{dt}$$

$$50 = (20\pi h - \pi h^2) \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{50}{20\pi h - \pi h^2}$$

$$h = 5$$

$$\frac{dh}{dt} = \frac{50}{75\pi}$$

$$= \frac{2}{3\pi}$$

$$\text{Now } \frac{ds}{dt} = \frac{ds}{dh} \times \frac{dh}{dt}$$

$$= (20\pi - 2\pi h) \times \frac{2}{3\pi}$$

$$h=5: \quad \frac{ds}{dt} = 10\pi \times \frac{2}{3\pi}$$

$$= \frac{20}{3} \text{ cm}^2/\text{sec.}$$

Question 7

$$a) \quad \int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx$$

$$u = x^2$$

$$x=0: u=0$$

$$\frac{du}{dx} = 2x$$

$$x = \frac{1}{\sqrt{2}}: u = \frac{1}{2}$$

$$du = 2x dx$$

$$= \frac{1}{2} \int_0^{\frac{1}{\sqrt{2}}} \frac{2x dx}{\sqrt{1-x^4}}$$

$$= \frac{1}{2} \int_0^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}}$$

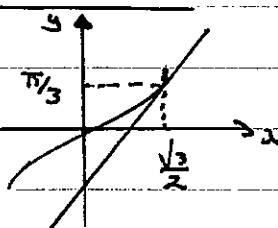
$$= \frac{1}{2} \left[\sin^{-1} u \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \left(\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{6} - 0 \right)$$

$$= \frac{\pi}{12}$$

b)



$$y = \sin^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\text{at } x = \frac{1}{2} : \frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{1}{4}}}$$

$$= \frac{1}{\sqrt{\frac{3}{4}}}$$

$$= 2$$

$\therefore$ equation of the tangent

$$y - \frac{\pi}{6} = 2 \left(x - \frac{1}{2} \right)$$

$$y - \frac{\pi}{6} = 2x - 1$$

$$y = 2x + \frac{\pi}{6} - 1$$

Cuts the 'y' axis at ϕ i.e. $x=0$

$$y = \frac{\pi}{6} - 1$$

$$\therefore \phi \left(0, \frac{\pi}{6} - 1 \right)$$

$$\therefore PQ = \sqrt{0^2 + \left(\frac{\pi}{6} - 1 - \frac{\pi}{6} \right)^2}$$

$$= \sqrt{1}$$

$$\text{Now } V = \text{Volume of a cone}$$

$$= \pi \int_0^1 x^2 dy$$

$$V = \frac{1}{3} \times \pi \times \left(\frac{\sqrt{3}}{2} \right)^2 \times \sqrt{3} - \pi \int_0^{\pi/3} \sin^2 y dy$$

$$= \frac{\pi \sqrt{3}}{4} - \frac{\pi}{2} \int_0^{\pi/3} 1 - \cos 2y dy$$

$$= \frac{\pi \sqrt{3}}{4} - \frac{\pi}{2} \left[y - \frac{\sin 2y}{2} \right]_0^{\pi/3}$$

$$= \frac{\pi \sqrt{3}}{4} - \frac{\pi}{2} \left[\left(\frac{\pi}{3} - \frac{\sin 2\pi/3}{2} \right) - 0 \right]$$

$$= \frac{\pi \sqrt{3}}{4} - \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$$

$$= \frac{\pi \sqrt{3}}{4} - \frac{\pi^2}{6} + \frac{\pi \sqrt{3}}{8}$$

$$= \frac{3\pi \sqrt{3}}{8} - \frac{\pi^2}{6}$$

$$= \frac{9\pi \sqrt{3}}{24} - \frac{4\pi^2}{24}$$

$$= \frac{\pi}{24} (9\sqrt{3} - 4\pi) \text{ cubic units}$$

$$\text{c) i) } g(x) = \frac{x+2}{x+1} \quad D: x < -1 \quad R: y < 1$$

$$g^{-1}(x): \quad x = \frac{y+2}{y+1}$$

$$xy + x = y + 2$$

$$xy - y = 2 - x$$

$$y(x-1) = 2-x$$

$$y = \frac{2-x}{x-1}$$

$$\therefore g^{-1}(x) = \frac{2-x}{x-1}$$

$$\therefore D: x < 1 \quad R: y < -1$$

ii) Curves will intersect on the line $y=x$:

$$\therefore \frac{x+2}{x+1} = x$$

$$x^2 + x = x + 2$$

$$x^2 = 2$$

$$x = \pm \sqrt{2}$$

$\therefore$ pt. of intersection $(-\sqrt{2}, -\sqrt{2})$

(as domain $g(x) < -1$)